AP PRECALCULUS

Unit 1 • Lesson 1.9 practice

Lesson 1.9 • Rational Functions and Vertical Asymptotes

5 multiple-choice questions and 1 free-response question

Directions

Questions 1 to 5 are multiple choice: choose the best of the four answer choices. Question 6 is free response: show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

Which line is a vertical asymptote of $r(x) = \dfrac{x+2}{x^2 - x - 6}$?

(A) $x = -2$

(B) $x = 3$

(C) $x = -2$ and $x = 3$

(D) $x = 6$

Question 2

NO CALCULATOR

$\lim\limits_{x \to 1^+} \dfrac{2x}{x-1}$ is

(A) $+\infty$

(B) $-\infty$

(C) 2

(D) 0

Question 3

NO CALCULATOR

$\lim\limits_{x \to -2^-} \dfrac{1}{(x+2)^2}$ is

(A) $-\infty$

(B) 0

(C) $\dfrac{1}{4}$

(D) $+\infty$

Question 4

NO CALCULATOR

Which function has a vertical asymptote at $x = 4$?

(A) $\dfrac{x-4}{x+4}$

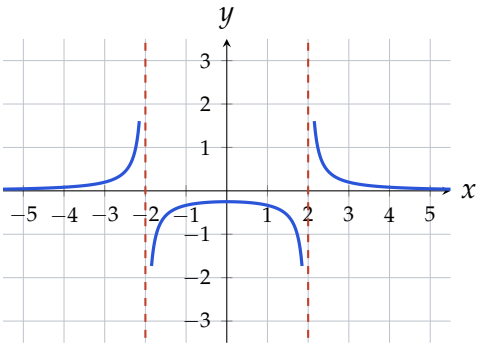
(B) $\dfrac{x^2 - 16}{x-4}$

(C) $\dfrac{}{(x-4)^2}$

(D) $\dfrac{(x-4)^2}{x}$

Question 5 (graph) NO CALCULATOR

The graph of $r(x) = \dfrac{1}{x^2 - 4}$ is shown, with vertical asymptotes $x = -2$ and $x = 2$. Which statement is true?



(A) $\lim_{x \to 2^-} r(x) = +\infty$
(B) $\lim_{x \to 2^+} r(x) = -\infty$
(C) $\lim_{x \to -2^+} r(x) = +\infty$
(D) $\lim_{x \to 2^-} r(x) = -\infty$

Question 6 (Free response) NO CALCULATOR

Let $r(x) = \dfrac{x - 1}{(x + 2)(x - 3)}$.

(a) Find the vertical asymptotes of the graph of r.
(b) For each vertical asymptote $x = a$, find $\lim\limits_{x \to a^-} r(x)$ and $\lim\limits_{x \to a^+} r(x)$. Show how you determine the signs.